Lea Brody-Heine

lea_brody-heine@alumni.brown.edu | Software Engineer | **BA & MSc**

Personal Website | LinkedIn | GitHub

Education

Brown University *BA* — Providence, RI
GPA: 4.0 — *Graduated: 05/23*

University of St Andrews *MSc Computer Science* — St Andrews, Scotland
GPA: First Class Honors with Distinction | Dean's List — *Graduated: 09/24*

Work Experience

Google Software Engineer — 12/24 – Current
Ads | Admob EngProd — *Sunnyvale, CA*

- Eliminated a 24-hour manual release bottleneck, saving $60K annually, by fully automating multiple manual release testing processes with 100% coverage.
- Took manual testing project from 0-to-1 in 9 months, authoring 4 technical design and proposal documents to architect end-to-end automation across Android and iOS.
- Delivered a new performance-metrics pipeline that contributed to 8,100+ lines of code, by driving a custom device-metrics crawler and parser from design to launch to automate Jank/Memory/latency tracking into our test result dashboard.
- Automated test-result analysis and stress testing across 5 ad formats (Banner, Interstitial, Native, App Open, and Rewarded), by building an agentic pipeline that autonomously analyzed logs and surfaced SDK breaking points.
- Implemented script that cut failing-test backlog by 50% (130 → 65) and processed 500+ bug reports by automating triage via Python scripts run daily.

Full Stack Web Developer — 09/24 – 10/24
AIM Executive Coaching | Paid Independent Contract — *Remote*

- Built and launched a fully responsive website (HTML, CSS, JavaScript, HubSpot), increasing discoverability 21% by implementing SEO best practices, optimizing load times, and developing for mobile, iPad, and desktop.
- Collaborated closely with the client to ensure alignment with wireframe specifications and user engagement goals, adapting design to reflect client feedback for a more personalized experience.

Master's Dissertation

Machine Learning for Pathology in Mast Cell Diseases — 01/24 – 08/24
University of St Andrews

- Constructed machine learning models to analyze tabular data and intestinal biopsy stains for mast cell disease research, employing generative AI to augment data and improve model training.
- Reduced processing time by 60% by leveraging a GPU PC, Docker, and containers to efficiently process complex programs.
- Increased detection accuracy by 25% by employing YOLOv8 and computer vision techniques to identify and count mast cells, spindle-shaped mast cells, and clusters in biopsy images.
- Enhanced research capabilities by augmenting over 200,000 data points (both tabular and image).
- Managed version control through GitHub to ensure reproducibility and collaborative development.
- Facilitated further research and reduced study time by 30% by designing user-friendly tools that employ both supervised and unsupervised models, allowing researchers and clinicians to input data to derive meaningful insights and discoveries and accelerate time to diagnosis.

Projects

AI & ML Development — 01/24 – 05/24
Personal & Collaborative Projects

- Engineered 5 logical agents using LogicNG and SAT4J Java libraries, implementing advanced strategies (SPS, SATS, PROBS) to solve complex decision-making problems.
- Developed and optimized a machine learning model for forecasting water pump status in Tanzania, achieving over 93% in prediction accuracy by leveraging scikit-learn, pandas, numpy, and Optuna for hyperparameter tuning.
- Implemented AI search algorithms for a flight route planner, optimizing pathfinding through DFS, BFS, A*, and SMA* algorithms, reducing search time by 35% using Euclidean distance heuristics.
- Delivered actionable insights by developing multiclass classification models for predicting cirrhosis patient outcomes, focusing on data imputation and handling unbalanced datasets, resulting in a 15% improvement in prediction accuracy.

Full Stack Web Development — 01/24 – 05/24
Personal & Collaborative Projects

- Produced a full stack web application with over 5,000 lines of code using a NoSQL database (MongoDB) and Node.js, Express, and Vue.js.
- Integrated RESTful APIs for API endpoints, optimized database management, and designed a full stack system architecture.
- Built a single-page web application for trivia quizzes, implementing 20+ interactive features using JavaScript, HTML, and CSS, enabling real-time question fetching and score tracking.

Technical Skills

Programming: Python, Java, JavaScript, HTML, CSS, TypeScript, SQL, Kotlin, Swift, Go/Golang

Development: Integration Testing, Performance Testing, Unit Testing, Release Engineering, Physical Device Testing, Large-Scale Test Infrastructure, Node.js, Express, Vue.js, React, Angular, RESTful APIs, MongoDB, MySQL, D3.js, Database Management (Relational and NoSQL), Firebase, Distributed Systems, On-Call Rotation, End-To-End Project Ownership

Machine Learning & AI: pandas, scikit-learn, TensorFlow, PyTorch, Keras, Deep Learning, LLM, Machine Learning Pipelines, Data Pipelines, Prompt Engineering, AI Fluency, Agentic Workflows, LLM Integration, Gemini API, AI-Assisted Development, Ethical AI Practices

Tools: Git, GitHub, Docker, Containers, Jupyter Notebook, Tableau, Batch Job Schedulers, Cloud Infrastructure, Linux

Practices & Specializations: Frontend, Backend, Full-Stack, SaaS, Mobile Development, Android, iOS, Object-Oriented Programming, Agile, Scrum, Microservices, Pair Programming, User-Centric Design, CI/CD, Release Automation, DevOps, On-Call Rotation, Bazel/bzl